Building Materials

Wei, Ya （魏亚）
Department of Civil Engineering

Tel.： +86-10-62771646
Email：yawei@tsinghua.edu.cn

Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood

Chapter Four：Concrete

1. Introduction
2. Compositions
3. Structure
4. Strength and Damage
5. Mix Proporationing
6. Early-age Properties
7. Volume Changes
8. Durability

4.2 Compositions

Types of Portland cements

According to ASTM standard, there are five basic types of Portland cement:

Type I regular cement, general use, called OPC

Type II moderate sulphate resistance, moderate heat of hydration, C_3A < 7%

Type III With increased amount of C_3S, high early strength

Type IV Low heat

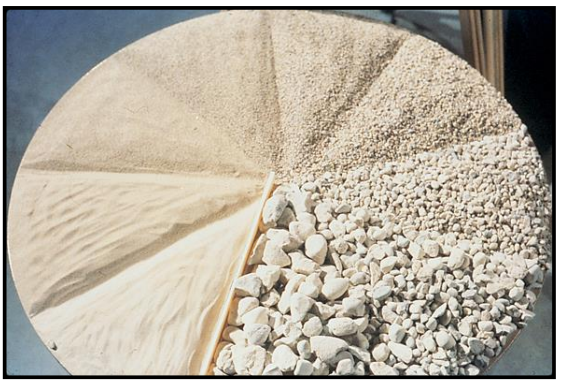
Type V High sulphate resistance

	I	II	III	IV	V
C_3S	50	45	60	25	40
C_2S	25	30	15	50	40
C_3A	12	7	10	5	4
C_4AF	8	12	8	12	10
$C\bar{S}H_2$	5	5	5	4	4

4.2 Compositions

Aggregate

- The aggregate occupies ~75% of the volume of concrete, so its quality is of great importance.



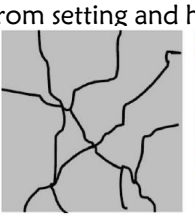
Range of particle sizes found in aggregate for use in concrete.

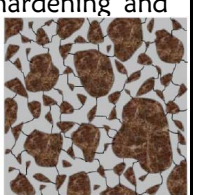
4.2 Compositions

Aggregate — Why add aggregate to concrete?

Aggregates have 3 main functions in concrete:

- To provide a mass of particles which are suitable to resist the action of applied loads and show better durability than cement paste alone.
- To provide a relatively cheap filler for the cementing material.
- To reduce volume changes resulting from setting and hardening and from moisture changes during drying.





Paste vs. concrete, which one is easier to develop cracks?

But:

- Cement must fill the space between aggregates
- Concrete must remain placeable

4.2 Compositions

Aggregate

The properties of concrete are affected by the properties of aggregate:

- The mineral character of aggregate affects the strength, durability, elasticity of concrete.
- The surface characteristics of aggregate affects the workability of fresh mass and the bond between the aggregate and cement paste in hardened concrete. If it is rough, workability decreases and bond increases.
- The grading of aggregate affects the workability, density and economy.



It is a common practice to use as much aggregate as possible in concrete!

4.2 Compositions

Aggregate

The important properties of aggregate are:

1) Shape and texture
2) Size gradation
3) Moisture content
4) Specific gravity
5) Unit weight



6) Durability and absence of deleterious materials

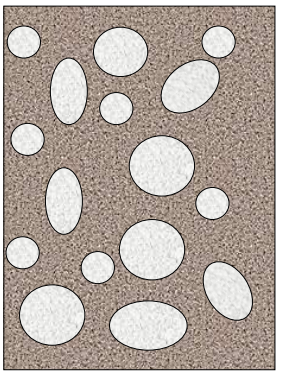
4.2 Compositions

Aggregate

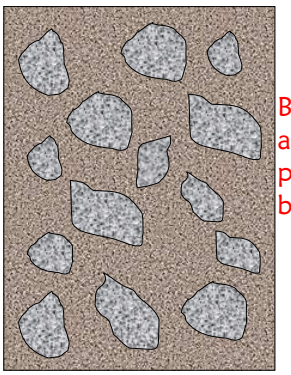
Shape and texture

Which one has better aggregate-paste bridging

Rounded



Crushed



Better aggregate-paste bridging

4.2 Compositions

Aggregate Shape and texture



Closeup of fine aggregate (sand)



Coarse aggregate. Rounded gravel (left) and crushed stone (right).



Lightweight aggregate. clay (left) and shale (right).

4.2 Compositions

Aggregate Particle shape

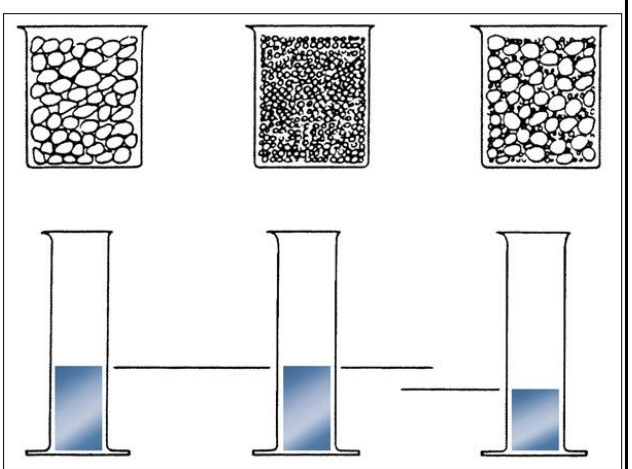
	Rounded		Crushed	
Shape	Spheroid	Flat / Oblong	Cubic	Flat / Oblong
				
Angularity	Rounded		Angular	
Surface state	Smooth		Rough	
Water demand	← Decreasing ←			
Workability, Pumpability	→ Decreasing →			

4.2 Compositions

Aggregate Size gradation

- Collection of various sizes in reducing the total volume of voids between aggregates

- In good concrete all aggregate particles must be covered by cement paste.



4.2 Compositions

Aggregate Size gradation

- Gradation is the particle-size distribution of an aggregate as determined by a sieve analysis.





Range of particle sizes found in aggregate for use in concrete.

4.2 Compositions

Aggregate Size gradation

- A good aggregate gradation for a particular concrete is the one that leads to a workable, dense and uniform concrete, without any segregation of particles.

- The grading are usually expressed as the percentage of material passing each sieve.

4.2 Compositions

Aggregate Size gradation

Workability: The ease with which a concrete mixture can be mixed, transported, placed in the form and compacted without any segregation.

Segregation: Seperation of the particles with different sizes and specific gravities.





The requirements of workability and absence of segregation tend to oppose each other. Thus, these two factors are interrelated. The major of these is workability which, in turn, affects most of the properties of concrete.

4.2 Compositions

Aggregate Size gradation

Determination of the Grading of Aggregate

There are two different methods for determining the agg. grading:

- Fineness Modulus (FM)
- Gradation Curve

- The fineness modulus (FM) of either fine or coarse aggregate is calculated by adding the cumulative percentages by mass retained on each of a specified series of sieves and dividing the sum by 100.

$$\text{Fineness Modulus (FM)} = \frac{\Sigma\ (\%\ \text{cumulative retained on each sieve})}{100}$$

4.2 Compositions

Aggregate Size gradation

- FM is an index of the fineness of an aggregate—the higher the FM, the coarser the aggregate.

For Fine Agg.→ #4, #8, #16, #30, #50, #100

For Coarse Agg.→Fine set+3/8"+3/4"+1 ½"+3"

The FM of the mixture of two or more agg. is the weighted average of the FM of that two or more agg.

Sieve size	
9.5 mm	(⅜ in.)
4.75 mm	(No. 4)
2.36 mm	(No. 8)
1.18 mm	(No. 16)
600 µm	(No. 30)
300 µm	(No. 50)
150 µm	(No. 100)
	Pan

4.2 Compositions

Aggregate Size gradation For Fine Agg.→ #4, #8, #16, #30, #50, #100

Sieve	Amount Retained on (g)	Amount Retained on (%)	% Cumulative Retained on
3/8"	0	0	0
#4	30	6	6
#8	80	16	22
#16	100	20	42
#30	120	24	66
#50	125	25	91
#100	35	7	98
Pan	10	2	100
Total=	500g		

$$FM = \frac{6+22+42+66+91+98}{100} = 3.25$$

Only standard sieves are included, if we were given #10 sieve you should not use that in calculations.

4.2 Compositions

For Fine Agg.→ #4, #8, #16, #30, #50, #100

Aggregate Size gradation For Coarse Agg.→Fine set+3/8”+3/4”+1 ½”+3”

Ex: Determine the FM for the 1000g sample of Coarse Agg.

Sieve	Amount Retained on (gr)	Amount Retained on (%)	% Cumulative Retained on
2"	70	7	7
1 1/2"	230	23	30
3/4"	350	35	65
3/8"	250	25	90
#4	100	10	100

$$FM = \frac{\text{Fine Set+3/8”+3/4”+1 ½”+3”}}{100}$$

$$FM = \frac{30+65+90+100+100+100+100+100+100}{100} = 7.85$$

4.2 Compositions

Aggregate Size gradation

Ex: The fine agg. with the FM=3.25 and the coarse agg. with the FM=7.85 are available. Combine them in such a way that the FM becomes 6.8

X : mass percentage of Fine agg.

$$\frac{3.25X+7.85(100-X)}{100} = 6.8 \longrightarrow X = 23$$

23% of fine agg. and 77% of coarse agg. should be mixed.

4.2 Compositions

Aggregate Size gradation

The FM is not always representative of the gradation of an aggregate sample and various gradation curves may give the same FM.

There are two different methods for determining the agg. grading:

- Fineness Modulus (FM)
- Gradation Curve

- In the gradation curves, the vertical axis represents the % passing and the horizontal axis represents the sieve opening.
- A logarithmic scale is used for horizontal axis.

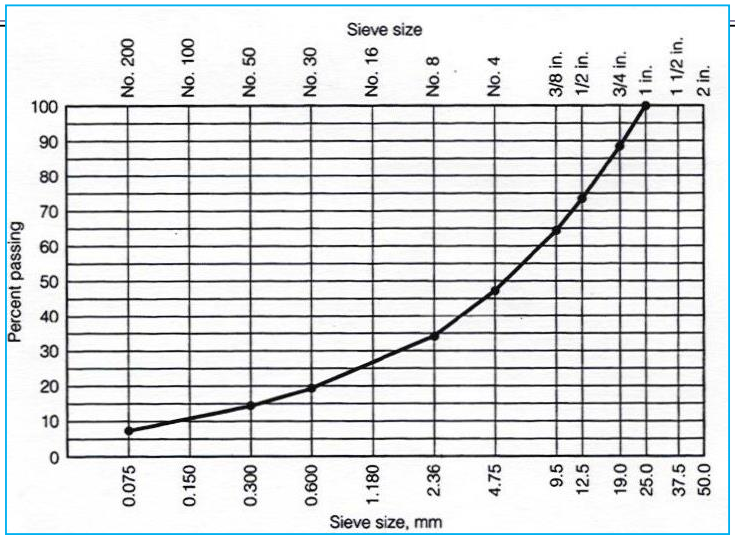
4.2 Compositions

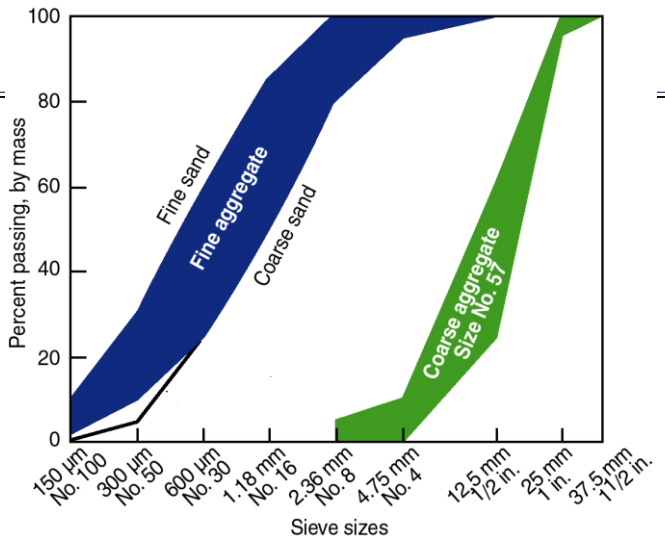
Aggregate

Sieve size		Percentage of individual fraction retained, by mass	Percentage passing, by mass	Cumulative percentage retained, by mass
9.5 mm	(⅜ in.)	0	100	0
4.75 mm	(No. 4)	2	98	2
2.36 mm	(No. 8)	13	85	15
1.18 mm	(No. 16)	20	65	35
600 µm	(No. 30)	20	45	55
300 µm	(No. 50)	24	21	79
150 µm	(No. 100)	18	3	97
	Pan	3	0	—
Total		100		283

Fineness modulus = 283 ÷ 100 = 2.83

Semi-log aggregate gradation chart



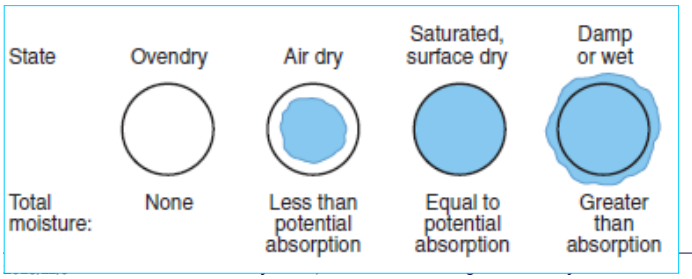


There is no single "ideal" grading curve. Instead, standards provide upper and lower limits.

4.2 Compositions

Aggregate **Moisture Content**

1. Ovendry — fully absorbent
2. Air dry — dry at the particle surface but containing some interior moisture, thus still somewhat absorbent
3. Saturated surface dry (SSD) — neither absorbing water from nor contributing water to the concrete mixture
4. Damp or wet — containing an excess of moisture on the surface (free water)



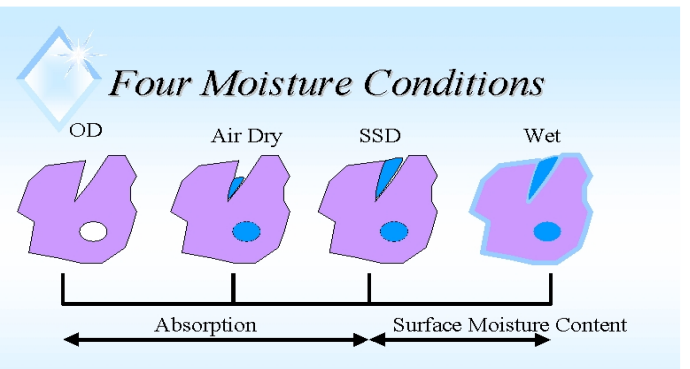


State in concrete?

25

4.2 Compositions

Aggregate **Moisture Content**



The amount of water added at the concrete batch plant must be adjusted for the moisture conditions of the aggregates in order to accurately meet the water requirement of the mix design.

4.2 Compositions

Aggregate Moisture Content

- SSD Condition → Equilibrium for Mositure Condition

1. If total moisture content = 0 → Agg. is bone-dry (oven dry)
2. If total moisture content < absorption capacity → It can absorb water
3. If total moisture content > absorption capacity → There is free water on the surface of agg.

- Mix Design Calculations are Based on Aggs in SSD Condition. Therefore, for aggs not being in that condition corrections have to be made
- w/c ratio → w should be "free water"

4.2 Compositions

Aggregate Moisture Content

$$\text{Abs.Cap} = \frac{W_{SSD}-W_{OD}}{W_{OD}}$$ (Absorption Capacity)

$$W_{OD} = \frac{W_{SSD}}{(1+\text{Abs.Cap.})}$$

$$\text{Moisture Content (m)} = \frac{W_{agg}-W_{OD}}{W_{OD}}$$

$$W_{agg} = W_{OD}\,(1+m)$$

4.2 Compositions

Aggregate Moisture Content

Ex: A sample of sand has the following properties:
Wet mass=625.2 g
Dry mass=589.9 g
Absorption=1.6%
Determine: (a)total moisture content, and(b)free moisture content

Solution:

Mass of water=625.2-589.9=35.3g

Total moisture content= $\frac{35.3}{589.9}\times 100 = 6.0\%$

Free moisture=6.0-1.6=4.4%

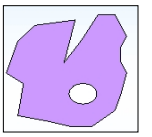
4.2 Compositions

Aggregate Specific Gravity (Relative Gravity) important for concrete mix design!

- The specific gravity of an aggregate is the mass of the aggregate in air divided by the mass of an equal volume of water.
- An aggregate with a specific gravity of 2.50 would thus be two and one-half times as heavy as water.
- Most natural aggregates have relative densities between 2.4 and 2.9 with corresponding particle (mass) densities of 2400 and 2900 kg/m^3 (150 and 181 lb/ft^3).
- Because the aggregate mass varies with its moisture content, specific gravity is determined at a fixed moisture content.

4.2 Compositions

Aggregate **Specific Gravity**



$$\text{Bulk Dry Sp.Gr.} = \frac{\textit{Dry}\ \text{Weight}}{(\textit{Total}\ \text{Particle Volume})\gamma_w} = \frac{W_s}{(V_s + V_i + V_p)\gamma_w}$$

$$\text{Bulk SSD Sp.Gr.} = \frac{\textit{SSD}\ \text{Weight}}{(\textit{Total}\ \text{Particle Volume})\gamma_w} = \frac{W_s + W_p}{(V_s + V_i + V_p)\gamma_w}$$

$$\textit{Apparent}\ \text{Sp.Gr.} = \frac{\textit{Dry}\ \text{Weight}}{(\textit{Volume}\ \text{Not Accessible to Water})\gamma_w} = \frac{W_s}{(V_s + V_i)\gamma_w}$$

Where:

W_s =weight of solids

V_s =volume of solids

V_i =volume of water impermeable voids

V_p =volume of water permeable voids

W_p =weight of water in the permeable voids when the aggregate is in the SSD condition

γ_w =unit weight of water

4.2 Compositions

Aggregate **Specific Gravity**

Determination of specific gravity and absorption of coarse aggregates:
a representative sample of the aggregate is soaked for 24 hours and weighed suspended in water. The sample is then dried to the SSD condition and weighed. Finally, the sample is dried to a constant weight and weighed.

$$\textit{Bulk}\ \text{Dry Sp.Gr.} = \frac{A}{B - C}$$

$$\textit{Bulk}\ \text{SSD Sp.Gr.} = \frac{B}{B - C}$$

$$\text{Apparent Sp.Gr.} = \frac{A}{A - C}$$

$$\text{Absorption}(\%) = \frac{B - A}{A}(100)$$

where
A=dry weight
B=SSD weight
C=submerged weight

4.2 Compositions

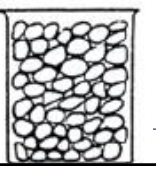
Aggregate **Unit Weight (Bulk density)**

- The bulk density or unit weight of an aggregate is the mass or weight of the aggregate required to fill a container of a specified unit volume. The volume referred to here is that occupied by both aggregates and the voids between aggregate particles.

- The approximate bulk density of aggregate commonly used in normal-weight concrete ranges from about 1200 to 1750 kg/m^3 (75 to 110 lb/ft^3).

- The void content between particles affects paste requirements in mix design.

- Angularity increases void content while improved grading decreases void content



4.2 Compositions

Admixtures

A material other than water, aggregate, cement, used as an ingredient of concrete or mortar added to the batch immediately before or during mixing.

There are two kinds of admixtures:

Chemical admixtures and Mineral admixtures.

Mineral Admixtures

- Pozzolanic (Low-calcium FA)
- Cementitious (Ground Granulated blast-furnace slag)
- Both pozzolanic and cementitious (High calcium FA)
- Natural materials (Volcanic Ash)

4.2 Compositions

Mineral Admixtures **Pozzolanic**

- A pozzolan is a material which, when combined with calcium hydroxide (lime), exhibits cementitious properties.
- Pozzolans are commonly used as an addition to Portland cement concrete mixtures to increase the long-term strength and other material properties of Portland cement concrete and in some cases reduce the material cost of concrete.
- – Examples are
  - Fly ash
  - Silica Fume
  - Rice Husk Ash
  - Metakaolin

4.2 Compositions

Mineral Admixtures **Pozzolanic reaction:**

Hydraulic reaction

OPC + water ⟶ C-S-H + **CH** + other hydrates

↓

Mainly responsible for strength
Impermeability of concrete

slag + water ⟶ C-S-H + other hydrates

Pozzolanic reaction

Pozzolan + **CH** + water ⟶ C-S-H + other hydrates

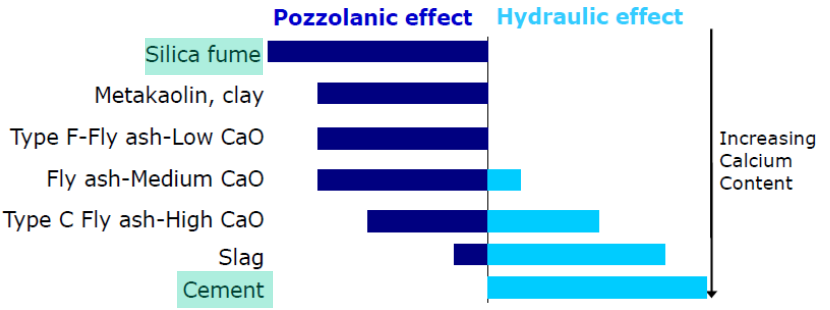
↓

More strength

Pozzolanic reaction converts the less dense CH phase and larger pores into denser CSH and smaller pore

4.2 Compositions

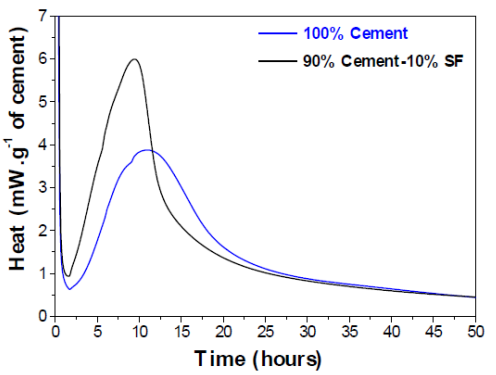
Mineral Admixtures Reactivity of SCMs



4.2 Compositions

Mineral Admixtures Reactivity

- SCM acts as nucleation sites for C-S-H growth.
- Enhances kinetics in acceleration period



4.2 Compositions

Mineral Admixtures Impact on hydration of cement:

1. Filler:
These additives/admixtures are finer than cement, so when added to concrete they occupy the small pores previously left vacant.

2. Nucleating:
Surface acts as nucleation site for hydrates.
These fine particles accelerate the rate of hydration and precipitation starts.

3. Pozzolanic:
it reduces the amount of CH and increase CSH, refine the pore structure and reduce the permeability of concrete thereby making it more resistant to the penetration of deleterious agents.

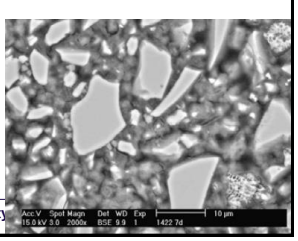
4.2 Compositions

Mineral Admixtures

Ground Granulated Blast Furnace Slag (GGBFS) or Slag



- A by-product of steel manufacture which is sometimes used as a substitute for Portland cement.
- It is a granular product, similar size and composition to Portland cement.
- highly cementitious in nature and, ground to cement fineness, hydrates like Portland cement.
- Blast furnace slag is blended with Portland cement clinker to form portland blast furnace slag cement.
- Low heat of hydration, slow strength gain, small pores, low permeability and high durability



4.2 Compositions

Mineral Admixtures Benefits of GGBFS

- **Durability**

GGBFS cement is routinely specified in concrete to provide protection against both sulphate attack and chloride attack.

- **Appearance**

The near-white color of GGBFS cement permits architects to achieve a lighter colour for exposed fair-faced concrete finishes, at no extra cost.

- **Strength**

Concrete containing GGBFS cement has a higher ultimate strength than concrete made with Portland cement





4.2 Compositions

Mineral Admixtures Fly Ash

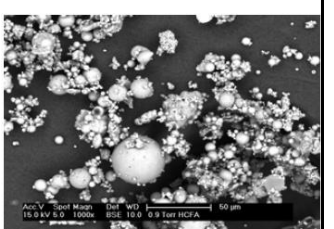
Fly ash is generally captured from the chimneys of coal-fired power plants; it has POZZOLANIC properties, includes substantial amounts of silicon dioxide (SiO_2) and calcium oxide (CaO).

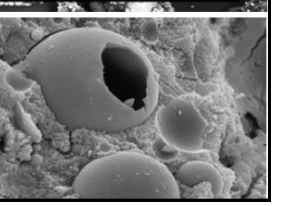
- Rounded particles, often hollow, of similar size to cement.

pozzolan reaction:

$Ca(OH)_2 + SiO_2 \rightarrow CSH$

- Slower reaction than cement, slower strength gain, lower heat of hydration.





4.2 Compositions

Mineral Admixtures Fly Ash

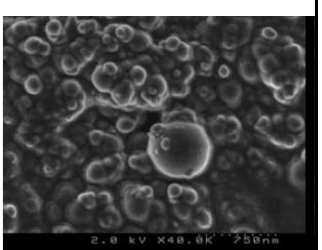
- improves its workability, reduces segregation, bleeding, heat evolution and permeability, inhibits alkali-aggregate reaction, and enhances sulfate resistance.

- Even though the use of fly ash in concrete has increased in the last 20 years, less than 20% of the fly ash collected was used in the cement and concrete industries.

- One of the most important fields of application for fly ash is PCC pavement, where a large quantity of concrete is used and economy is an important factor in concrete pavement construction.

4.2 Compositions

Mineral Admixtures Silica Fume



- Waste from silicon production (now expensive!), such as semiconductor industry
- consists of very fine particles with a surface area ranging from 13,000 to 30,000 m^2 /kg.
- a highly effective pozzolanic material, in addition to optimum filler.
- improves compressive strength, and abrasion resistance; reduces permeability of concrete to chloride ions; and therefore helps in protecting reinforcing steel from corrosion, especially in chloride-rich environments such as coastal regions.
- Used in concrete with special requirements:

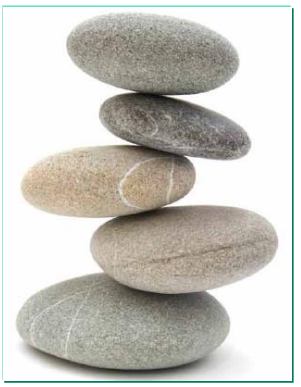
- high strength concrete
- low permeability

4.2 Compositions

Mineral Admixtures Rice Husk Ash

- This is a bio waste from the husk left from the grains of rice. It is used as a pozzolanic material in cement to increase durability and strength.
- The silica is absorbed from the ground and gathered in the husk where it makes a structure and is filled with cellulose. When cellulose is burned, only silica is left which is grinded to fine powder which is used as pozzolan.





QUESTIONS?